Calorimetry

→ It is study of heat for liquids and solids.

→ Liquids and solids have no substantial change in volume on heating
⇒ $W = 0$.

→ ΔQ is responsible to increase temperature or phase change.

→ Unit of heat = Joule, Calorie ; Dimension = $[ML^2T^{-2}]$

1 calorie: It is defined as amount of heat required by 1gm of water to increase temperature from 14.5°C to 15.5°C at 1atm pressure

$$1 \text{ calorie} = 4.18 \text{ Joules} \approx 4.2 \text{ Joules}$$
$$1 \text{ Kcal} = 1000 \text{ cal} = 4180 \text{ J} \approx 4200 \text{ J}$$

Q A boy of mass 60kg is running with speed 36km/hr. Find his kinetic energy in calorie.

$V = 36 \text{ km/hr} = 10 \text{ m/s}$

$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 60 \times (10)^2$

$KE = 3000 \text{ J}$

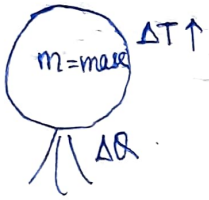
$4.2 \text{ J} = 1 \text{ cal}$

$\Rightarrow 1 \text{ J} = \frac{1}{4.2} \text{ cal}$

$\Rightarrow 3000 \text{ J} = \frac{3000}{4.2} \text{ cal}$

Specific Heat Capacity / Specific Heat (s):

→ Heat required per unit mass per unit change in temperature



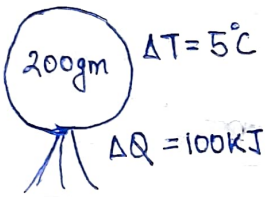
$$S = \frac{\Delta Q}{m\Delta T}$$

$$\text{Unit} = \frac{\text{Joule}}{\text{kg-kelvin}}$$

$$\text{Dimension} = \frac{ML^2T^{-2}}{MK} = [L^2T^{-2}K^{-1}]$$

→ Property of object

Q



Find S.

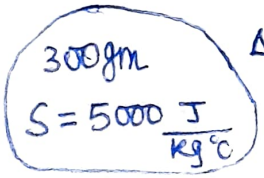
$$S = \frac{\Delta Q}{m\Delta T} = \frac{100000 \text{ J}}{0.2\text{kg} \times 5 \text{ kelvin}} = 100000 \frac{\text{J}}{\text{kg-kelvin}}$$

Note:

$$\boxed{\Delta Q = mS\Delta T}$$

* If $S \neq \text{const} \Rightarrow \Delta Q = \int dQ = m\int s\,dT$

Q



Find heat required.

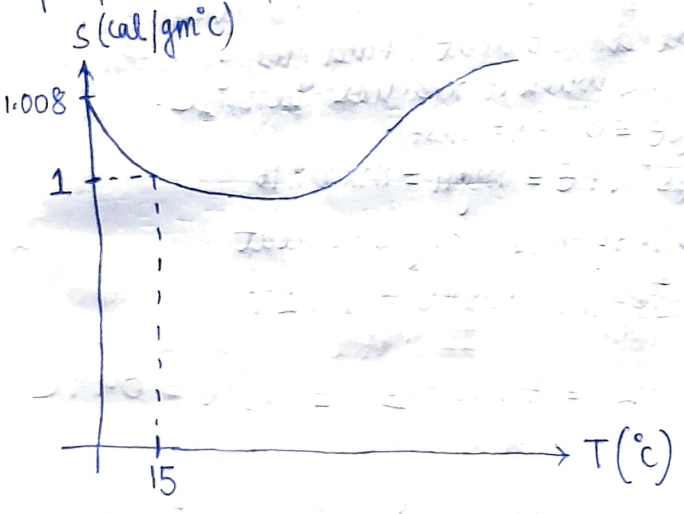
$$\Delta Q = mS\Delta T = 0.3\text{kg} \times 5000 \frac{\text{J}}{\text{kg°C}} \times 5°\text{C} = 7500 \text{ J}$$

Q (2kg) Tinitial = 100 kelvin, $S = (1000 + 2T)\frac{J}{kg\text{-}K}$ Find heat required to increase temperature by 100 kelvin

$$\Delta Q = \int dQ = m\int s\,dT = 2kg\int_{100}^{200}(1000+2T)\,dT\ \frac{J}{kg} = 2\left(1000T + T^2\right)_{100}^{200}$$

$$\Delta Q = 2\left((1000\times200 + 200^2) - (1000\times100 + 100^2)\right)\ J$$

Specific heat of water:



$$S_{water} = \frac{1\,cal}{gm\,^\circ c} = \frac{4.2\,J}{gm\,^\circ c} = \frac{1\,kcal}{kg\,^\circ c} = \frac{4200\,J}{kg\,^\circ c}$$

$$S_{ice} = S_{steam} = \frac{S_{water}}{2} = \frac{0.5\,cal}{gm\,^\circ c} = \frac{2.1\,J}{gm\,^\circ c} = \frac{500\,cal}{kg\,^\circ c} = \frac{2100\,J}{kg\,^\circ c}$$

Heat capacity (C):



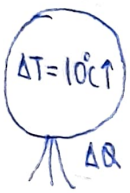
→ It is heat required by object per unit change in temperature.

$$C = \frac{\Delta Q}{\Delta T}$$ unit = J/kelvin, Dimension = $[ML^2T^{-2}K^{-1}]$

→ $C = \frac{\Delta Q}{\Delta T} = \frac{ms\Delta T}{\Delta T} = ms$

- → depends on object
- → depends on quantity of object

Q



Find heat capacity of this object.

$$C = \frac{\Delta Q}{\Delta T} = \frac{4000\,cal}{10^\circ C} = \frac{400\,cal}{^\circ C} = \frac{400\times4.2\,J}{^\circ C}$$

Q Find heat capacity of

① 3kg water ⇒ $C = ms = 3kg \times \frac{1\,kcal}{kg\,^\circ C} = \frac{3\,Kcal}{^\circ C} = \frac{3000\,cal}{^\circ C} = \frac{3000\times4.2\,J}{^\circ C}$

② 12kg water ⇒ $C = 12kg \times \frac{1\,kcal}{kg\,^\circ C} = \frac{12\,Kcal}{^\circ C} = \frac{12000\,cal}{^\circ C}$

③ 6gm water ⇒ $C = 6gm \times \frac{1\,cal}{gm\,^\circ c} = \frac{6\,cal}{^\circ C}$

Mechanical Equivalent of Heat :

→ Heat and work are same.

Equation :

Heat = const × Work

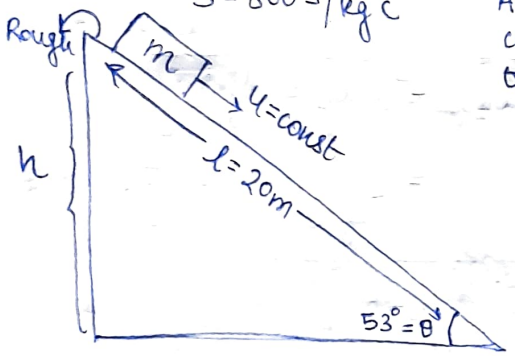
Work = const × Heat

$$\boxed{W = JH}$$

4.2 Joules = J × 1 calorie

$$\boxed{J = 4.2 \frac{\text{Joules}}{\text{Calorie}}} \Rightarrow \text{Mechanical equivalent of Heat}$$

Q



$S = 800\ J/kg^\circ C$

Assume that 70% of mechanical energy is converted into heat. Find change in temp. of block when it reaches the base.

$\Delta KE = 0$ {v=const}

Loss in PE = $mgH = m \times 10 \times 16$

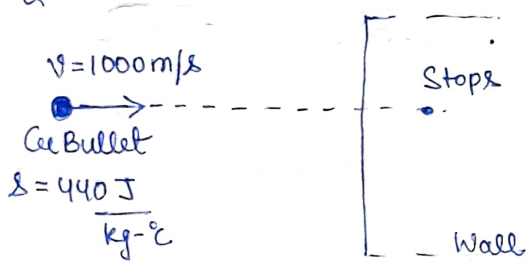
70% of this changes to heat

$\Rightarrow \frac{70}{100} \times m \times 10 \times 16 = m S \Delta T$

$\Delta T = \frac{7 \times 10 \times 16}{10 \times 800} = \left(\frac{7}{50}\right)^\circ C = 0.14^\circ C$

$h = 20 \sin 53^\circ = 16m$

Q



60% loss in TME is converted into heat. Find increase in temp. of Cu bullet

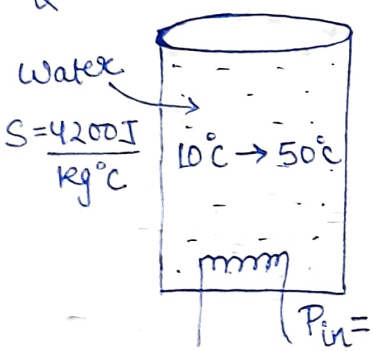
$TME = \frac{1}{2} m v^2$

60% of TME gets lost in heat

$\frac{60}{100} \times \frac{1}{2} m v^2 = m S \Delta T$

$\Delta T = \frac{10^6 \times 6}{2 \times 440 \times 10} = \frac{3}{44} \times 10^4\ ^\circ C$

Q



Find how many kgs of water can be heated in 1 hour by this geyser to raise temp. from 10°C to 50°C.

Pout in 1 sec = 800W

Pout in 1 hr = $800 \frac{J}{s} \times 60 \times 60\ s$

Geyser efficiency = 80%

$\eta = \frac{P_{out}}{P_{in}} \times 100 \Rightarrow 80 = \frac{P_{out}}{1000W} \times 100$

$P_{out} = 800W$

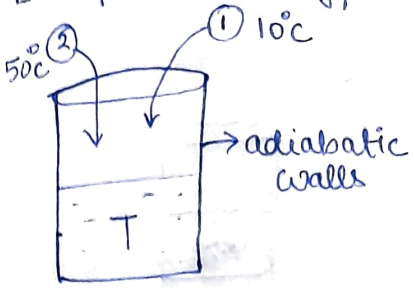
So,

$\Delta Q = 800\ J \times 60 \times 60 = m S \Delta T$

$= 800 \times 60 \times 60 = m \times \frac{4200}{kg^\circ} \times 40^\circ$

$m = \frac{800 \times 60 \times 60}{4200 \times 40}\ kg = \frac{240}{14}\ kg = \frac{120}{7}\ kg$

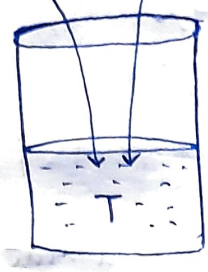
Law of Calorimetry / Law of mixture:



Heat lost by ② = Heat gained by ①

$$m_2 s_2 (50 - T) = m_1 s_1 (T - 10)$$

Ex m_1, s_1, T_1 liq ① $\quad m_2, s_2, T_2$ liq ②



$(T_1 > T_2)$

Find final temperature of mixture.

Method ①:

Heat lost by liq ① = Heat gained by liq ②

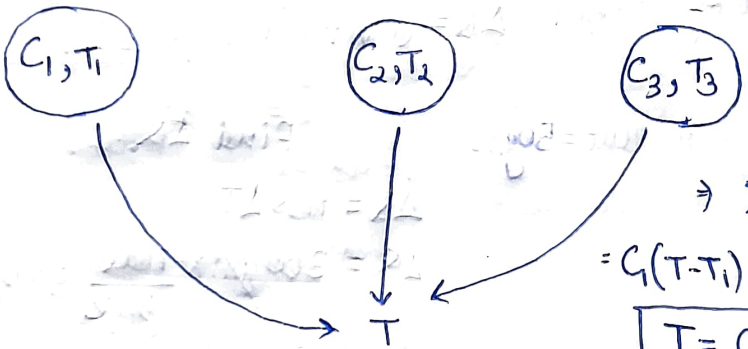
$$m_1 S_1 (T_1 - T) = m_2 S_2 (T - T_2)$$

$$T = \frac{m_1 S_1 T_1 + m_2 S_2 T_2}{m_1 S_1 + m_2 S_2} = \frac{C_1 T_1 + C_2 T_2}{C_1 + C_2}$$ (C = heat capacity)

Method ②

$$\boxed{\Sigma \Delta Q_{gain} = 0} \Rightarrow m_1 s_1 (T - T_1) + m_2 s_2 (T - T_2) = 0$$

Ex

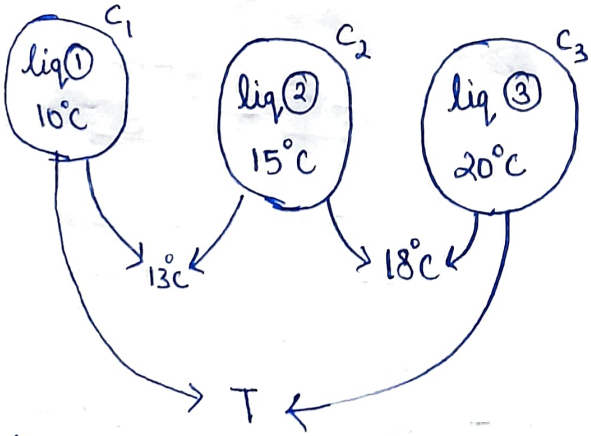


Find final temp. T

$\Rightarrow \Sigma \Delta Q_{gain} = 0$

$= C_1(T - T_1) + C_2(T - T_2) + C_3(T - T_3) = 0$

$$\boxed{T = \frac{C_1 T_1 + C_2 T_2 + C_3 T_3}{C_1 + C_2 + C_3}}$$



Find T.

On mixing liq ① and ②

$C_1 \times 3 = C_2 \times 2$

$$\boxed{\frac{C_1}{C_2} = \frac{2}{3}} \quad -①$$

On mixing liq ② and ③

$C_2 \times 3 = C_3 \times 2$

$$\boxed{\frac{C_2}{C_3} = \frac{2}{3}} \quad -②$$

On mixing liq ① and liq ③

$C_1 (T - 10) = C_3 (20 - T)$

$$\frac{C_1}{C_3} = \frac{20 - T}{T - 10} = \frac{4}{9} \Rightarrow T = \left(\frac{190}{13}\right)^{\circ}C$$

Multiplying ① and ②

$$\frac{C_1}{C_3} = \frac{4}{9}$$

Water Equivalent:

→ It is mass of water which has same heat capacity as that of object.

→ Unit = kg

→ Dimension = [M]





Q



$S = 420 \frac{J}{kg°C}$

Find its water equivalent.

$$C = ms = 2kg \times 420 \frac{J}{kg°C} = m_w \times 4200 \frac{J}{kg°C}$$

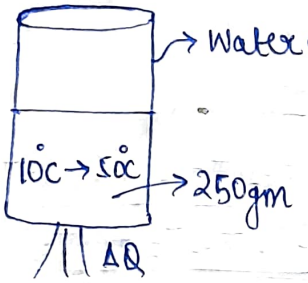
$$\boxed{m_w = 0.2kg}$$

#Note:



$\Delta Q = \Delta Q_w + \Delta Q_c$, ie heat provided by geyser can heat water as well as the container. If water equivalent of container is given then we can replace container with that amount of water, $\Delta Q = (m_w + \text{water eq.}) s \Delta T$

Q



Find ΔQ.

$\Delta Q = ms\Delta T$

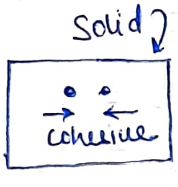
$\Delta Q = 300 gm \times \frac{1 cal}{gm°C} \times 40°C$

$\Delta Q = 12000 cal$

Phase Change:

- Solid $\underset{\text{freezing}}{\overset{\text{melting}}{\rightleftarrows}}$ liquid
- liquid $\xrightarrow{\text{boiling}}$ Vapour

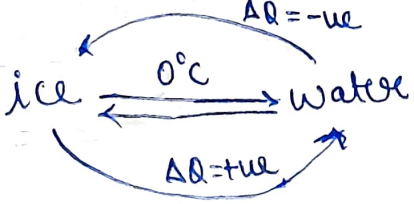
Melting



As T↑, KE of molecules ↑, Thermal agitations begins.

At T = $T_{melting\ point}$, molecules break the cohesive force

In this case, heat is used for phase change.



Latent heat of fusion (L_f):

→ Amount of heat given to solid per unit mass to convert it into liquid at melting point is defined as L_f.

→ unit = J/kg

→ Dimension = $[L^2T^{-2}]$

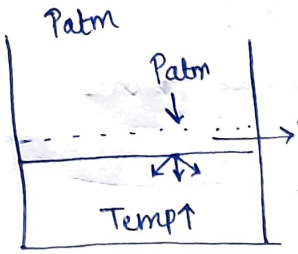
Solid (m) $\xrightarrow[\Delta Q]{M.P.}$ liquid (m)

$$L_f = \frac{\Delta Q}{m}$$

$$\Rightarrow \Delta Q = mL_f$$

For ice ⟶ water $L_f = 80 \frac{cal}{gm} = 80 \frac{Kcal}{kg}$

Vapourization:



P_{vp} = Vapour Pressure

When $T = T_{B.P} \Rightarrow P_{atm} = P_{vp} \Rightarrow$ Boiling starts.

Latent heat of vapourization:

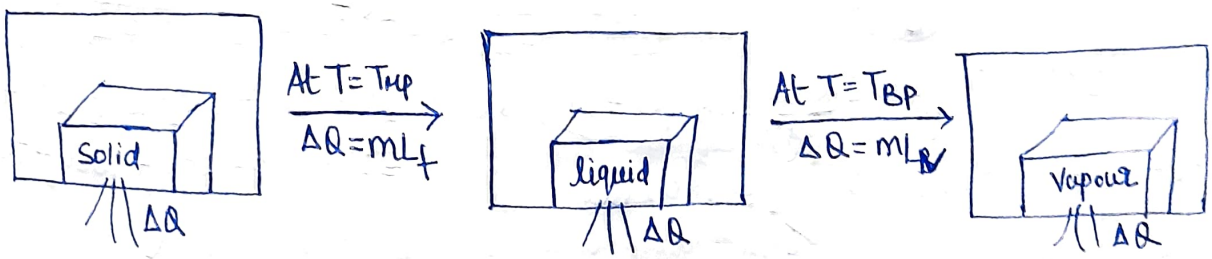
→ It is amount of heat required per unit mass to convert liquid to vapours at boiling point.

$$L_v = \frac{\Delta Q}{m}$$

liquid (m) $\xrightarrow[\Delta Q]{B.P.}$ Vapour (m)

For water $\xrightarrow{100^\circ C}$ Vapour (steam) $L_v = 540 \frac{cal}{gm} = 540 \frac{Kcal}{kg}$

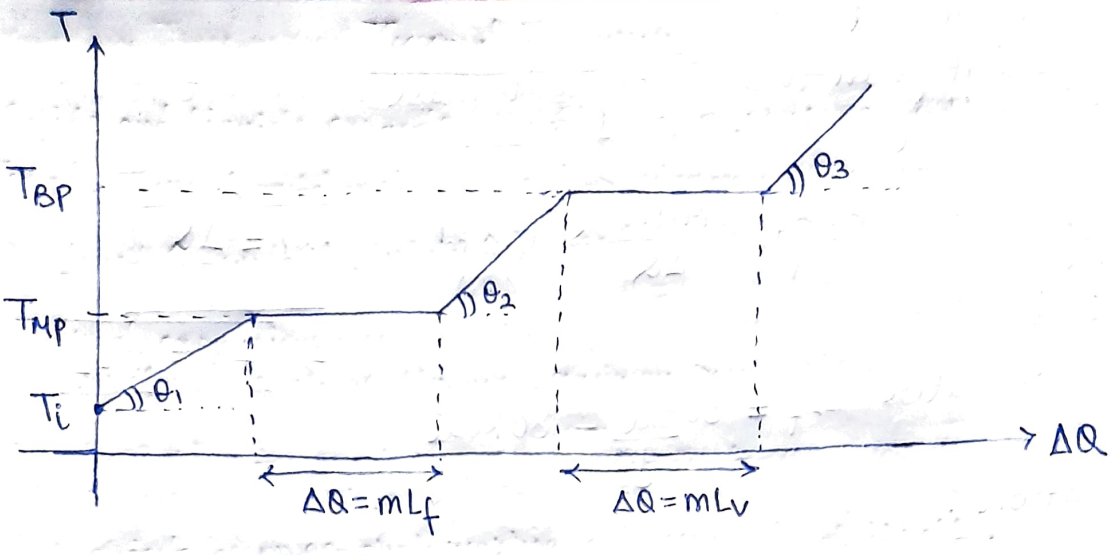
#Note:



$\Delta Q = mS_s\Delta T$ $\Delta Q = mS_l\Delta T$ $\Delta Q = mS_v\Delta T$

*On providing heat, temperature raises. When temp. reaches melting pt/ Boiling pt., phase change occurs. Temperature during phase change remains constant and heat provided is used for phase change.

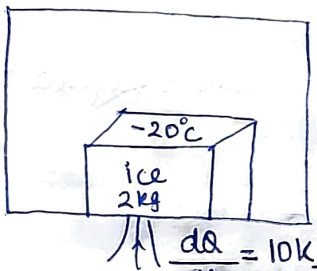
Graph on next page.



$$\tan\theta_1 = \frac{1}{mS_s} \qquad \tan\theta_2 = \frac{1}{mS_\ell} \qquad \tan\theta_3 = \frac{1}{mS_v}$$

$$\boxed{\tan\theta_1 : \tan\theta_2 : \tan\theta_3 = \frac{1}{S_s} : \frac{1}{S_\ell} : \frac{1}{S_v}}$$

Q



① Find heat required to convert this ice into 2kg steam at 150°C

Given: $S_{ice} = S_{steam} = 0.5\frac{kcal}{kg°C}$

$S_{water} = 1\frac{kcal}{kg°C}$

$L_f = 80\frac{kcal}{kg}$ $\qquad L_v = 540\frac{kcal}{kg}$

Ice -20°C $\xrightarrow{\Delta Q_1}$ Ice 0°C $\xrightarrow{\Delta Q_2}$ W 0°C $\xrightarrow{\Delta Q_3}$ W 100°C $\xrightarrow{\Delta Q_4}$ S 100°C $\xrightarrow{\Delta Q_5}$ S 150°C

$\Delta Q_1 = mS\Delta T = 2kg \times 0.5\frac{kcal}{kg°C} \times 20°C = 20kcal$

$\Delta Q_2 = mL_f = 2kg \times 80\frac{kcal}{kg} = 160kcal$

$\Delta Q_3 = mS\Delta T = 2 \times 1 \times 100 = 200kcal$

$\Delta Q_4 = mL_v = 2 \times 540 = 1080kcal$

$\Delta Q_5 = mS\Delta T = 2 \times 0.5 \times 50 = 50kcal$

$\therefore \Delta Q = \Delta Q_1 + \Delta Q_2 + \Delta Q_3 + \Delta Q_4 + \Delta Q_5$

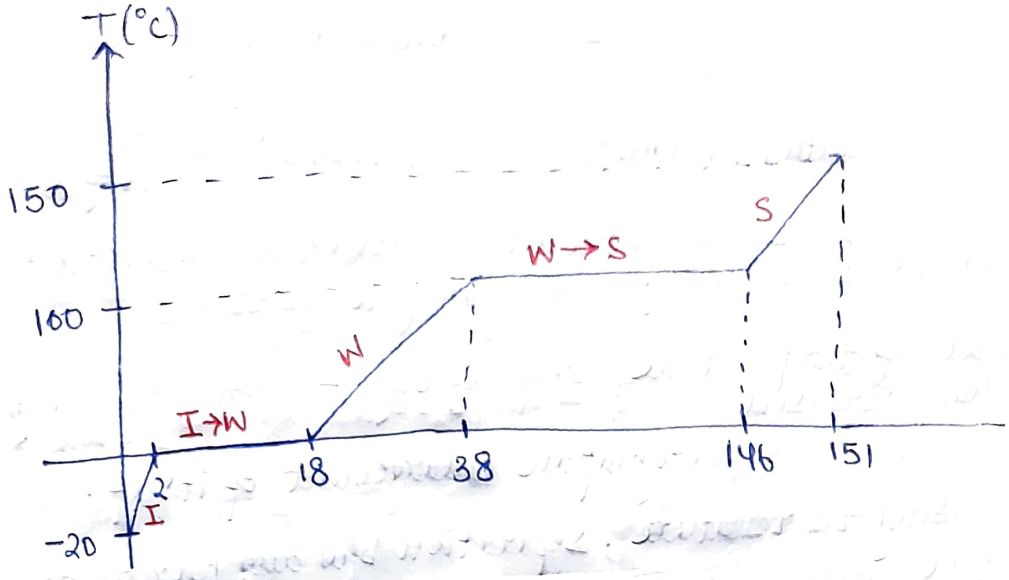
$= 20 + 160 + 200 + 1080 + 50$

$\Delta Q = 1510kcal$

② Find total time taken.

$$t = \frac{1510kcal}{10\frac{kcal}{sec}} = 151\,sec$$

③ Draw T(°C) v/s time (in sec).



*** Q 2kg Ice at -20°C is mixed with 1kg steam at 140°C. Find final temp and composition.

Soln: First we find the amount of heat to change 1kg steam at 140°C to 1kg ice at -20°C

1kg steam at 140°C ⟶ 1kg ice at -20°C

$\Delta Q = 1kg \times 0.5\frac{kcal}{kg°C} \times 40°C + 1kg \times 540\frac{kcal}{kg} + 1\times1\times100 + 1kg \times 80\frac{kcal}{kg} + 1\times0.5\times20$

$\Delta Q = 20 + 540 + 100 + 80 + 10 = 750\,kcal$

∴ 1kg steam at 140°C − 750kcal ⟶ 1kg ice at -20°C

⇒ 1kg steam at 140°C = 1kg ice at -20°C + 750kcal ——— (1)

Now, we have

2kg Ice at -20°C + 1kg steam at 140°C

= 2kg Ice at -20°C + 1kg ice at -20°C + 750kcal

= 3kg Ice at -20°C + 750kcal

Now we need to calculate, at what temp. we can reach using 750kcal heat.

So, For ice at 0°C ⇒ $\Delta Q = 3\times0.5\times20 = 30\,kcal$

= 3kg Ice at 0°C + 720kcal

Now, to change ice to water at 0°C ⇒ $\Delta Q = mL_f = 3\times80 = 240\,kcal$

= 3kg water at 0°C + 480kcal

Now, with 480kcal heat we can raise temp upto, $480 = 3\times1\times\Delta T \Rightarrow \Delta T = 160$.

But there will be phase change at 100°C.

So, For water at 100°C ⇒ $\Delta Q = 3\times1\times100 = 300\,kcal$

= 3kg water at 100°C + 180 kcal

Now, to change 3kg water to steam at 100°C we need heat = $mL_v = 3\times540 = 1620\,kcal$.

But we have only 180kcal. So mass of water that can turn into steam using 180 kcal

$180 = m\times540 \Rightarrow m = 1/3\,kg$

$\frac{1}{3}$kg steam at 100°C + $\frac{8}{3}$ kg water at 100°C ⇒ final ans.